

SECTION 1: BASIC DEFINITIONS & TYPES (Q1–Q20)

1. Define solution in chemistry.

A **solution** is a **homogeneous mixture** of two or more substances in which the components are uniformly distributed at the molecular level.

2. What is a binary solution? Give one example.

A **binary solution** is a solution containing **two components only**: a solute and a solvent.

Example: Sugar in water.

3. Define solute and solvent with an example from liquid–liquid solution.

- **Solute:** The component present in smaller amount.
- **Solvent:** The component present in larger amount.

Example: In ethanol–water solution, **ethanol = solute**, **water = solvent**.

4. What is meant by homogeneous mixture? How is it related to a solution?

A **homogeneous mixture** has **uniform composition** throughout.

A **solution** is always a **homogeneous mixture**.

5. Classify the following:

- (a) **Air** → Gaseous solution
 - (b) **Brass** → Solid solution
 - (c) **Sugar in water** → Liquid solution
-

6. What are solid solutions? Give two examples.

Solutions where **solvent is solid**.

Examples: Brass (Cu–Zn), Bronze (Cu–Sn)

7. What are liquid solutions? Give two examples.

Solutions where **solvent is liquid**.

Examples: Salt in water, Alcohol in water

8. What are gaseous solutions? Give two examples.

Solutions where **solvent is gas**.

Examples: Air, Moist air

9. Define aqueous and non-aqueous solutions.

- **Aqueous solution:** Solvent is water (NaCl in water)
 - **Non-aqueous solution:** Solvent other than water (I₂ in alcohol)
-

10. Saturated solution

A solution containing **maximum amount of solute** at a given temperature.

11. Unsaturated solution

Contains **less solute than maximum possible** at given temperature.

12. Supersaturated solution

Contains **more solute than saturated solution**, unstable.

13. Ideal solution

A solution that **obeys Raoult's law** over entire concentration range.

14. Non-ideal solution

A solution that **deviates from Raoult's law**.

15. Two characteristics of ideal solution

- $\Delta H_{\text{mix}} = 0$
 - $\Delta V_{\text{mix}} = 0$
-

16. Two characteristics of non-ideal solution

- $\Delta H_{\text{mix}} \neq 0$
 - $\Delta V_{\text{mix}} \neq 0$
-

17. Miscible liquids

Liquids that mix in all proportions.

Example: Alcohol and water

18. Immiscible liquids

Liquids that do not mix.

Example: Oil and water

19. Gas dissolved in solid

Type: Solid solution

Example: Hydrogen in palladium

20. Solid dissolved in solid

Type: Solid solution

Example: Brass

SECTION 2: CONCENTRATION TERMS (Q21–Q55)

21. Mass percentage (% w/w)

$$\% (w/w) = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

22. Volume percentage (% v/v)

$$\% (v/v) = \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$$

23. Mass by volume (% w/v)

$$\% (w/v) = \frac{\text{Mass of solute (g)}}{\text{Volume of solution (mL)}} \times 100$$

24. Mole fraction

$$X_A = \frac{n_A}{n_A + n_B}$$

25. Molarity (M)

$$M = \frac{\text{Moles of solute}}{\text{Volume of solution (L)}}$$

Unit: mol L⁻¹

26. Molality (m)

$$m = \frac{\text{Moles of solute}}{\text{Mass of solvent (kg)}}$$

Unit: mol kg⁻¹

27. SI unit of mole fraction

No unit (dimensionless)

28. Temperature-dependent

Molarity (volume changes with temperature)

29. Meaning of 10% (w/w) NaCl

10 g NaCl in 100 g solution.

30. Mass % of urea

$$= \frac{10}{10 + 90} \times 100 = 10\%$$

31. Mole fraction of water

Moles glucose = 18/180 = 0.1

Moles water = 180/18 = 10

$$X_{water} = \frac{10}{10.1} = 0.99$$

32. Molality

$$m = \frac{0.5}{0.5} = 1\ m$$

33. Molarity

$$M = \frac{0.2}{1} = 0.2\ M$$

34. Molarity vs Molality

Molarity

Molality

Depends on volume

Depends on mass

Temp dependent

Temp independent

35. Reason

Volume changes with temperature, mass does not.

36. Mole fraction of A

$$X_A = \frac{2}{5} = 0.4$$

37. Mole fraction of ethanol

$$X = \frac{5}{20} = 0.25$$

38. Relationship

$$X_{solute} + X_{solvent} = 1$$

39. Moles present

$$n = M \times V = 0.2 \times 0.25 = 0.05 \text{ mol}$$

40. Mass of NaOH

$$n = 0.1 \times 0.5 = 0.05$$

$$\text{Mass} = 0.05 \times 40 = 2 \text{ g}$$

41. ppm

Parts per million; used for very dilute solutions.

42. 0.9% saline

0.9 g per 100 mL

43. 1 molal glucose

180 g glucose in 1000 g water

44. Normality

$$N = \frac{\text{Gram equivalents}}{\text{L solution}}$$

Used in acid-base and redox reactions

45. Effect on molarity

Decreases on heating.

46. Effect on molality

No change

47. Dilute solutions

$M \approx m$

48. Molarity

$$n = \frac{4}{40} = 0.1$$
$$M = \frac{0.1}{0.5} = 0.2 M$$

49. Dilution

$$M_1 V_1 = M_2 V_2$$
$$V_1 = \frac{0.5 \times 500}{2} = 125 \text{ mL}$$

50. Stock solution

Highly concentrated solution used to prepare dilute solutions.

51. Advantage of mole fraction

Temperature independent.

52. Mole percent

$$\text{\text{Mole \%}} = X \times 100$$

53. Conversion method

Use density $\rightarrow$ mass $\rightarrow$ moles $\rightarrow$ molarity.

54. Dilution

Lowering concentration by adding solvent.

$$M_1 V_1 = M_2 V_2$$

55. Mass %

Advantage: Temperature independent

Disadvantage: Not useful for reactions

SECTION 3: SOLUBILITY & HENRY'S LAW (Q56–Q85)

56. Solubility

Maximum amount of solute dissolved in given solvent at given T.

57. Units

g per 100 g solvent

58. Factors

- Temperature
 - Nature of solute/solvent
-

59. Temperature effect

Solubility of solids **increases** with temperature.

60. Gas solubility vs pressure

Directly proportional.

61. Henry's law

Solubility $\propto$ pressure.

62. Expression

$$p = K_H x$$

63. Henry's constant

Measure of solubility; large $K_H \rightarrow$ low solubility.

64. Pressure doubled

Solubility doubles.

65. Temperature effect

Gas solubility **decreases** with increase in temperature.

66. Aerated water

More CO₂ dissolves at low temperature.

67. Bends

Excess N₂ dissolved under high pressure.

68. Soda bottle

Sudden pressure drop → CO₂ escapes violently.

69. Fish die

Oxygen solubility decreases in hot water.

70. Soft drinks

CO₂ packed under high pressure.

71. Diving

He–O₂ mixture reduces nitrogen solubility.

72. Large KH

Low solubility.

73. Small KH

High solubility.

74. Limitation

Fails at high pressure.

75. Deviation

High pressure or reactive gases.

76. CO₂ vs O₂

CO₂ reacts with water → higher solubility.

77. Nature of gas

Polar gases dissolve more (NH₃).

78. Higher solubility

Lower KH.

79. Organic solvents

Similar polarity to gases.

80. Units of KH

atm, bar

81. Pressure over solution

Pressure exerted by gas above liquid.

82. Colligative properties

ΔT_f , π

83. Salting out

NaCl reduces soap solubility.

84. Common ion effect

Reduced solubility due to common ion.

85. Influence

Solubility decreases.

SECTION 4: RAOULT'S LAW (Q86–Q115)

86. Raoult's law (non-volatile solute)

$$p = X_{\text{solvent}} p^0$$

87. Expression

$$p_A = X_A p_A^0$$

88. Total vapour pressure

$$p = X_A p_A^0 + X_B p_B^0$$

89. Ideal solution

Obeys Raoult's law.

90. Characteristics

$$\Delta H_{\text{mix}} = 0, \Delta V_{\text{mix}} = 0$$

91. Non-ideal solution

Deviates from Raoult's law.

92. Positive deviation

Observed vapour pressure > ideal.

93. Negative deviation

Observed vapour pressure < ideal.

94. Examples

Positive: Ethanol–water

Negative: Acetone–chloroform

95. Azeotrope

Constant boiling mixture.

96. Difference

Minimum bp: positive deviation

Maximum bp: negative deviation

97. Examples

Min: Ethanol–water

Max: HCl–water

98. Ideal solution

$$\Delta H_{\text{mix}} = 0, \Delta V_{\text{mix}} = 0$$

99. Positive deviation

$$\Delta H_{\text{mix}} > 0, \Delta V_{\text{mix}} > 0$$

100. Negative deviation

$$\Delta H_{\text{mix}} < 0, \Delta V_{\text{mix}} < 0$$

101. Graph

Curve above Raoult line.

102. Graph

Curve below Raoult line.

103. Forces

Strong A–B interactions.

104. Forces

Weak A–B interactions.

105. Partial vapour pressure

Pressure by one component.

106. Expression

$$p_A = X_A p_A^0$$

107. Vapour mole fraction

$$y_A = \frac{p_A}{p_{total}}$$

108. Dalton's law

Total pressure = sum of partial pressures.

109. Relation

$$p = p_A + p_B$$

110. Raoult vs Henry

Raoult is special case when solute = solvent.

111. Explanation

Gases are dilute; liquids require similar interactions.

112. Nearly ideal

Very dilute solutions.

113. Similar liquids

Similar intermolecular forces.

114. Ethanol–water

Hydrogen bonds broken → positive deviation.

115. Acetone–chloroform

Strong H-bonding → negative deviation.

SECTION 5: COLLIGATIVE PROPERTIES (Q116–Q150)

116. Colligative properties

Depend only on number of particles.

Examples: ΔT_b , ΔT_f

117. Meaning

“Colligative” = collective effect.

118. Four properties

- Lowering vapour pressure

- Elevation bp
 - Depression fp
 - Osmotic pressure
-

119. Relative lowering

$$\frac{p^0 - p}{p^0}$$

120. Elevation bp

Increase in boiling point.

121. Depression fp

Decrease in freezing point.

122. Osmotic pressure

Pressure to stop osmosis.

123. Relation

$$\frac{p^0 - p}{p^0} = X_{solute}$$

124. BP increases

Lower vapour pressure.

125. FP decreases

Solvent crystallization hindered.

126. ΔT_b

$$\Delta T_b = K_b m$$

127. Kb

Depends on solvent only.

128. ΔT_f

$$\Delta T_f = K_f m$$

129. Kf

Depends on solvent.

130. van't Hoff factor

Number of particles formed.

131. Definition

$$i = \frac{\text{Observed}}{\text{Calculated}}$$

132. Electrolytes

Dissociate into ions.

133. Non-electrolytes

No dissociation.

134. Osmotic pressure

$$\pi = CRT$$

135. SI unit

Pascal (Pa)

136. Advantage

Works for very dilute solutions.

137. Hypertonic

Cell shrinks (crenation).

138. Hypotonic

Cell swells.

139. Isotonic

Normal saline in IV fluids.

140. Association detection

Lower colligative effect than expected.

141. Dissociation detection

Higher colligative effect.

142. Higher ΔT_f

Solute dissociates.

143. Vapour pressure

Always decreases.

144. Expression

$$\frac{p^0 - p}{p^0} = \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{solvent}}}$$

145. Reason

Very dilute solutions.

146. Methods

Ebulliometry, Cryoscopy

147. Degree of dissociation

$$i = 1 + \alpha(n - 1)$$

148. Valid only for dilute

Assume ideal behavior.

149. Best conditions

Dilute, non-electrolyte solutions.

150. Conclusion

They contain equal total number of moles.